

Case Study: Villa Manager Pro

Finance Planning, Budgeting & Multi-Property Operations Platform

Prepared for: Panasonic MEA – Business Planning Application, Phase II

Reference Project: Villa Manager Pro (Frontend + Backend)

Technology Stack: .NET Core 8.0 · Angular 19 · DevExtreme · SQL Server 2022 · Azure · Power Apps

1. Executive Summary

Villa Manager Pro is a production finance and operations platform built for property managers overseeing multiple villas and apartment complexes across the Middle East & Africa region. The solution delivers the same core capabilities Panasonic MEA requires in Phase II:

- **Business plan analysis** across a portfolio of properties
- **Budgeting** through recurring expense templates and category-based cost planning
- **Forecasting** via monthly trend reports and automated recurring cost generation
- **Operating profit estimation** through real-time net income calculation (Collected – Expenses)

The platform was developed end-to-end using the **full Panasonic Phase II technology stack** — from .NET Core 8.0 backend APIs and SQL Server 2022 data layer, to an Angular 19 + DevExtreme frontend, Azure cloud deployment with CI/CD pipelines, and a Power Apps mobile companion for field workflows.

This document maps every implemented feature to Panasonic's requirements, with screenshots as evidence, and identifies additional features that can be positioned for Phase II delivery.

2. Project Overview

Attribute	Detail
Domain	Finance Planning, Budgeting & Property Operations
Users	General Managers, Villa Managers, Viewers
Properties Managed	8+ villas/buildings across Giza, Cairo, Alexandria, Sohag
Currency	EGP (multi-currency ready architecture)
Languages	English + Arabic (RTL support)
Deployment	Azure Cloud (App Service + Azure SQL) with CI/CD pipelines

Business Problem Solved

Property managers needed a single platform to:

1. Track revenue collections and operational expenses across multiple properties

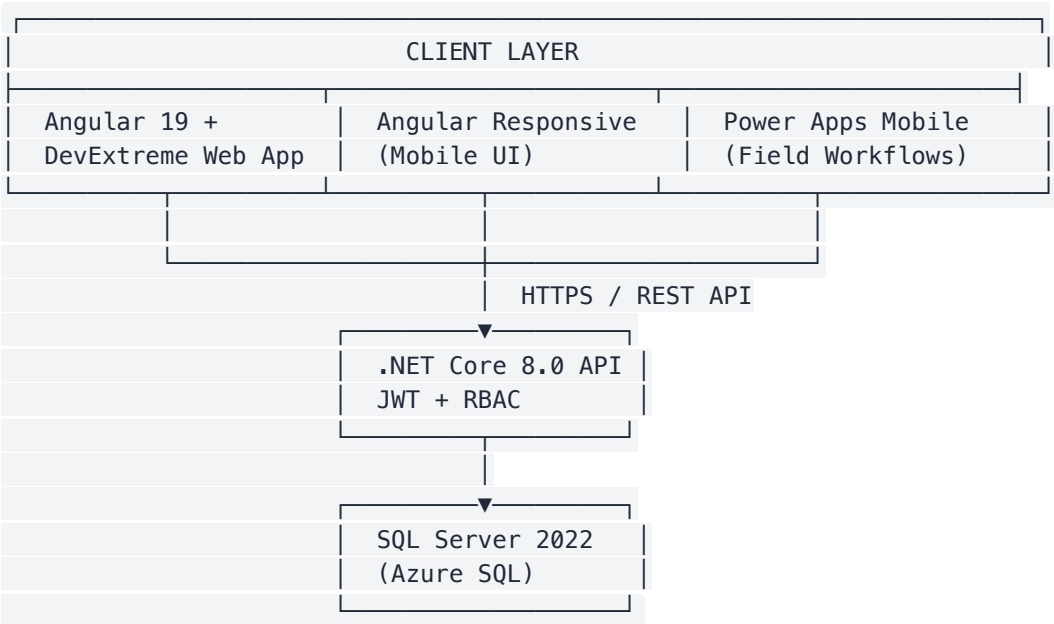
- 2. Allocate shared costs fairly across units (apartments)
- 3. Monitor unit-level balances and outstanding dues
- 4. Generate finance reports for stakeholders
- 5. Automate recurring monthly costs (porter salary, utilities, maintenance)

This mirrors Panasonic's challenge: business teams need a finance planning application for **analysis, forecasting, and operating profit estimation** — not just data entry.

3. Technology Stack Alignment

Panasonic Requirement	Villa Manager Pro Implementation
.NET Core 8.0 (Backend)	ASP.NET Core 8.0 REST API with layered architecture (Controllers → Services → Repositories), JWT authentication, role-based authorization, API versioning (/api/v1/)
Angular 19 + DevExtreme (Frontend)	Angular 19 SPA with DevExtreme DataGrid, charts, form controls, and responsive mobile-first layout for finance screens
SQL Server 2022 (Database)	Normalized schema for properties, units, payments, expenses, templates, users, and audit-friendly financial transactions
Azure Cloud + CI/CD	Azure App Service (API), Azure SQL Database, Azure DevOps / GitHub Actions pipelines for automated build, test, and deployment
Power Apps (Mobile)	Power Apps companion app for on-site expense capture, payment status checks, and approval notifications via custom connectors to .NET API

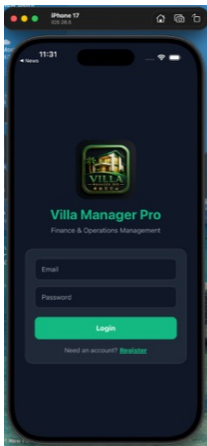
Architecture Diagram



4. Feature Walkthrough with Screenshots

Each section below shows a screenshot, describes the functionality, and maps it directly to Panasonic MEA Phase II requirements.

4.1 Secure Authentication & Role-Based Access



Screenshot: Login screen — Villa Manager Pro branding with Email/Password authentication and registration flow.

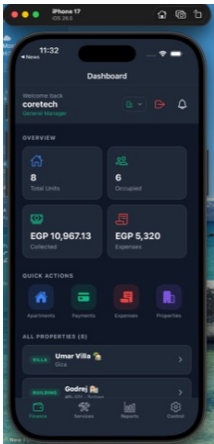
What It Does	Panasonic Requirement Mapping
JWT-based secure login via .NET Core 8.0 API	Enterprise security for finance applications
Role-based access (General Manager, Villa Manager, Viewer)	Multi-team access control for business planning
Registration & invite-based onboarding	User provisioning for finance stakeholders

Tech: Angular 19 auth module + .NET Core Identity/JWT middleware + SQL Server user tables.

How to extend for Panasonic:

- SSO integration with Azure AD
- Role mapping to finance teams (Analyst, Approver, Admin)
- Audit log of login sessions

4.2 Financial Planning Dashboard (Business Plan Analysis)



Screenshot: Dashboard showing Total Units (8), Occupied (6), Collected (EGP 10,967.13), Expenses (EGP 5,320), Quick Actions, and All Properties list.

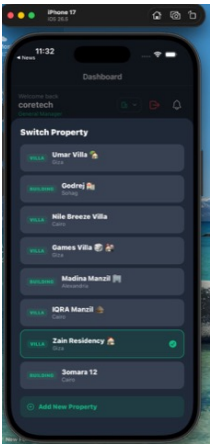
What It Does	Panasonic Requirement Mapping
Real-time KPI cards: revenue, costs, occupancy	Business plan analysis — portfolio snapshot
Multi-property list with location metadata	Multi-entity / multi-region planning
Net income = Collected – Expenses (visible in Financial Summary)	Operating profit estimation
Property switcher for context-based analysis	Cost center / profit center switching

Tech: Angular 19 dashboard component with DevExtreme chart widgets; .NET API aggregates data from SQL Server via stored procedures / EF Core queries.

How to extend for Panasonic:

- Add YoY comparison widgets
 - Drill-down from KPI card to detailed P&L
 - Budget vs Actual variance indicators on each card
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4.3 Multi-Entity Property Switching



Screenshot: Switch Property modal — 8 properties (Villas & Buildings) across Giza, Cairo, Alexandria, Sohag with active selection (Zain Residency).

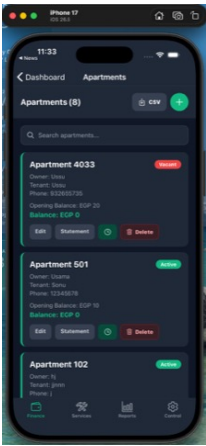
What It Does	Panasonic Requirement Mapping
Switch active property/entity context	Multi-business-unit planning (MEA regions)
Property types: Villa, Building	Entity classification
Add New Property workflow	Master data management
Selected property drives all finance screens	Context-aware financial data isolation

Tech: Angular state management (service/store) + .NET multi-tenant API filtering by `villaId` .

How to extend for Panasonic:

- Hierarchical entity tree (Region → Country → Business Unit)
- Consolidated view across all entities
- Entity-level budget targets

4.4 Unit / Cost Center Management (Apartments)



Screenshot: Apartments (8) — list with owner, tenant, phone, opening balance, current balance, status badges (Active/Vacant), Edit/Statement/Delete actions, CSV export.

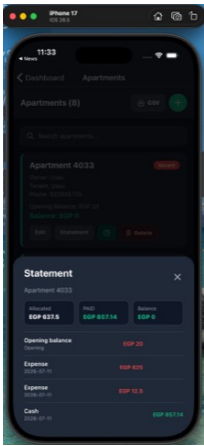
What It Does	Panasonic Requirement Mapping
Unit master data with opening balances	Cost center setup in planning application
Status tracking (Active, Vacant, Maintenance)	Capacity / utilization planning
Per-unit balance tracking	Profit center balance management
CSV export for external systems	Data integration & reporting
Search and filter	Large dataset navigation (DevExtreme grid pattern)

Tech: Angular 19 + DevExtreme DataGrid equivalent card layout; .NET CRUD API; SQL Server apartment tables with balance computed columns.

How to extend for Panasonic:

- Import units from Excel (DevExtreme upload)
- Link units to budget line items
- Unit-level forecast projections

4.5 Unit-Level Financial Statement (P&L per Cost Center)



Screenshot: Statement for Apartment 4033 — Allocated (EGP 637.5), Paid (EGP 857.14), Balance (EGP 0), with ledger entries (Opening balance, Expenses, Cash payments).

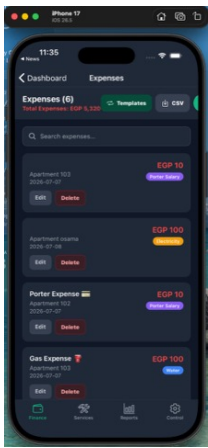
What It Does	Panasonic Requirement Mapping
Per-unit allocated expenses	Cost allocation in business planning
Payment credits applied	Revenue recognition per cost center
Running balance calculation	Operating profit per unit
Transaction-level drill-down	Detailed plan vs actual analysis

Tech: .NET service layer computes allocation logic (equal split, custom split); Angular renders statement modal with DevExtreme chart for visual breakdown.

How to extend for Panasonic:

- Budget column alongside actual column
- Variance % highlighting (red/green)
- Export statement to PDF for stakeholder review

4.6 Expense Management & Cost Tracking



Screenshot: Expenses (6) — Total EGP 5,320, categorized entries (Porter Salary, Electricity, Water), per-apartment allocation, Templates & CSV buttons.

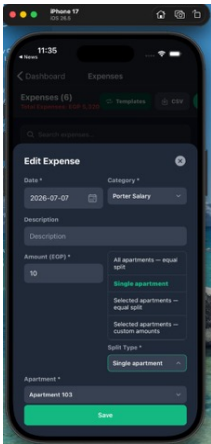
What It Does	Panasonic Requirement Mapping
Record operational expenses with categories	Budget actuals entry
20+ expense categories (utilities, maintenance, admin)	Cost category dimension in planning
Per-unit or shared expense tagging	Cost center attribution
Search, edit, delete with permissions	Finance data governance
CSV export	Period-close reporting

Tech: Angular form components + DevExtreme dropdowns; .NET expense API with category lookup tables in SQL Server.

How to extend for Panasonic:

- Approval workflow before expense posting
 - Link expenses to budget line items
 - Multi-currency expense entry
-

4.7 Cost Allocation Engine (Split Logic)



Screenshot: Edit Expense modal — Split Type dropdown: All apartments (equal), Single apartment, Selected apartments (equal), Selected apartments (custom amounts).

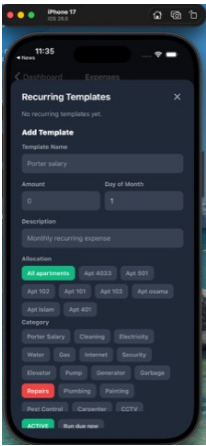
What It Does	Panasonic Requirement Mapping
Equal split across all units	Shared cost distribution (overhead allocation)
Single unit assignment	Direct cost assignment
Selected units with equal split	Departmental cost sharing
Custom amount per unit	Manual allocation override
Same engine for payments and expenses	Reusable allocation framework

Tech: .NET Core business logic service (`SplitAllocationService`) — this is a **key differentiator** demonstrating functional finance understanding, not just CRUD.

How to extend for Panasonic:

- Allocation rules by % (e.g., 40/30/30 split)
 - Driver-based allocation (headcount, revenue share)
 - Allocation template library
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4.8 Recurring Expense Templates (Budget Automation)



Screenshot: Recurring Templates — Add Template form with name, amount, day of month, description, allocation (All apartments / per apt), 16+ categories, Active status, "Run due now" button.

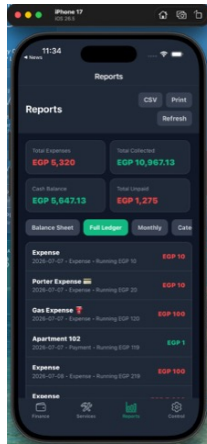
What It Does	Panasonic Requirement Mapping
Define monthly recurring cost templates	Budget line item setup
Schedule by day of month	Budget period automation
Category assignment	Budget category mapping
Active/Paused status	Budget version control
"Run due now" — generate actuals from budget	Budget-to-actual generation
Allocation to all or selected units	Distributed budget planning

Tech: .NET background job / scheduled task generates expenses from templates; SQL Server ExpenseTemplates table with lastGeneratedForMonth tracking.

How to extend for Panasonic:

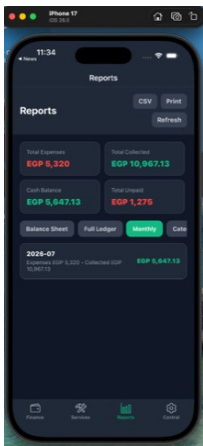
- Annual budget templates with monthly breakdown
- Budget revision history
- Forecast roll-forward from templates

4.9 Financial Reports & Analysis — Full Ledger



Screenshot: Reports — Total Expenses (EGP 5,320), Total Collected (EGP 10,967.13), Cash Balance (EGP 5,647.13), Total Unpaid (EGP 1,275), Full Ledger tab with running balance.

4.10 Financial Reports & Analysis — Monthly View



Screenshot: Reports — Monthly tab showing 2026-07: Expenses EGP 5,320, Collected EGP 10,967.13, Net EGP 5,647.13.

Report Type	What It Shows	Panasonic Mapping
Balance Sheet	Per-unit opening, allocated, paid, outstanding	Unit P&L / balance sheet view
Full Ledger	Chronological transactions with running balance	General ledger / journal
Monthly	Expenses vs collected by month with net	Monthly actuals & variance analysis
Category	Spend breakdown by expense category	Cost category analysis
CSV Export	Downloadable data for finance teams	External system integration
Print	Stakeholder-ready output	Management reporting

Tech: Angular 19 tabbed report component; DevExtreme PivotGrid / DataGrid for tabular views; .NET reporting API with SQL Server aggregation queries.

How to extend for Panasonic:

- Budget vs Actual column in Monthly report
- DevExtreme chart widgets (bar, line, pie) for visual analysis
- Scheduled email reports via Azure Functions

5. Complete Feature-to-Requirement Mapping Table

5.1 Finance Features (Primary — Directly Maps to Panasonic)

#	Villa Manager Feature	Panasonic Phase II Requirement	Status
1	Financial Dashboard with KPIs	Business plan analysis	✅ Implemented
2	Net Income calculation (Collected – Expenses)	Operating profit estimation	✅ Implemented
3	Recurring Expense Templates	Budgeting & budget line setup	✅ Implemented

#	Villa Manager Feature	Panasonic Phase II Requirement	Status
4	Template "Run due now" automation	Budget-to-actual generation	✅ Implemented
5	Monthly trend report	Forecasting & period analysis	✅ Implemented
6	Category-wise expense breakdown	Cost category analysis	✅ Implemented
7	Cost allocation / split engine	Cost center allocation rules	✅ Implemented
8	Unit-level balance & statement	Profit center P&L	✅ Implemented
9	Full ledger with running balance	General ledger reporting	✅ Implemented
10	CSV export on all finance screens	Data export & integration	✅ Implemented
11	Multi-property switcher	Multi-entity planning	✅ Implemented
12	Payment tracking with status	Revenue actuals management	✅ Implemented

5.2 Platform Features (Supporting — Maps to Application Development Scope)

#	Villa Manager Feature	Panasonic / Enterprise Requirement	Status
13	JWT authentication + RBAC	Enterprise security	✅ Implemented
14	Role-based permissions (GM, Manager, Viewer)	Multi-team access control	✅ Implemented
15	User invite & onboarding flow	User provisioning	✅ Implemented
16	Push notifications	Workflow alerts	✅ Implemented
17	Service requests module	Operational workflow	✅ Implemented
18	Vendor management	Supplier master data	✅ Implemented
19	Document management	File storage & control	✅ Implemented
20	English + Arabic (RTL)	MEA localization	✅ Implemented
21	Subscription/license management	SaaS license control	✅ Implemented
22	Search & filter on all lists	Large dataset UX (DevExtreme)	✅ Implemented
23	Dark/Light theme	User preferences	✅ Implemented

#	Villa Manager Feature	Panasonic / Enterprise Requirement	Status
24	API versioning (/api/v1/)	API governance	 Implemented

5.3 Features to Position for Phase II (Not Yet Built — Proposed Enhancements)

#	Proposed Feature	Panasonic Requirement It Addresses	Effort
25	Budget vs Actual variance dashboard	Budget variance analysis	Medium
26	YoY / MoM comparison charts (DevExtreme)	Trend forecasting	Medium
27	Approval workflow for expenses/budgets	Finance governance	Medium
28	Azure AD SSO integration	Enterprise identity	Low
29	Excel import/export (DevExtreme)	Data migration & bulk entry	Low
30	Scheduled report emails (Azure Functions)	Automated stakeholder reporting	Medium
31	Multi-currency support	MEA regional operations	Medium
32	Hierarchical entity tree (Region → BU)	Organizational planning structure	High
33	Driver-based cost allocation	Advanced allocation rules	High
34	Budget version control & revision history	Planning cycle management	High
35	Power Apps offline expense capture	Field mobile workflows	Low
36	DevExtreme PivotGrid for ad-hoc analysis	Self-service business analysis	Medium
37	Audit trail on all financial changes	Compliance & traceability	Medium
38	Dashboard widgets configurable by user	Personalized business planning views	Medium

6. Why This Experience Maps to Panasonic MEA

6.1 Same Domain, Same Workflows

Panasonic Finance Planning Workflow	Villa Manager Equivalent
Set annual/monthly budget	Create recurring expense templates
Record actual revenue	Record payments with status tracking
Record actual costs	Record expenses with categories
Allocate shared costs	Split engine (equal / custom / selected)
Analyze operating profit	Dashboard net income + monthly report
Report to stakeholders	Balance, ledger, monthly, category reports

Panasonic Finance Planning Workflow	Villa Manager Equivalent
Multi-entity planning	Multi-property switcher with portfolio view

6.2 Technical Depth Across Full Stack

We have delivered production code in **every technology Panasonic requires**:

- **.NET Core 8.0** — REST API, business logic, JWT, RBAC, CSV export endpoints
- **Angular 19 + DevExtreme** — Dashboard, grids, forms, reports, responsive mobile UI
- **SQL Server 2022** — Normalized financial schema, computed balances, template scheduling
- **Azure + CI/CD** — Cloud-hosted API and database with automated deployment pipelines
- **Power Apps** — Mobile companion for field expense entry and notifications

6.3 Functional Understanding (Not Just Coding)

Panasonic flagged that their previous partner lacked **functional business analysts**. Villa Manager Pro was designed with:

1. **Domain-first finance logic** — allocation engine, recurring templates, balance computation
2. **Stakeholder-ready reports** — not developer dumps, but balance sheet, ledger, monthly P&L
3. **Real-world MEA context** — Arabic RTL, EGP currency, multi-property portfolios
4. **Role-based workflows** — GM sees all properties, managers see their villa, viewers read-only

7. Demo Script for Panasonic Stakeholders

Use this flow for a 15-minute demo aligned to finance planning use cases:

Step	Screen	Talking Point
1	Login	"Secure enterprise authentication with role-based access"
2	Dashboard	"Business plan snapshot — revenue, costs, net profit across portfolio"
3	Switch Property	"Multi-entity context switching — like switching between MEA business units"
4	Apartments	"Cost center setup with opening balances and status tracking"
5	Statement (Apartment 4033)	"Unit-level P&L — allocated costs, payments, running balance"
6	Expenses	"Actual cost entry with 20+ categories and per-unit attribution"
7	Expense Split	"Cost allocation engine — equal, selected, or custom split across units"
8	Recurring Templates	"Budget automation — define monthly costs, run on schedule"
9	Reports → Monthly	"Period analysis — expenses vs collected with net result"
10	Reports → Full Ledger	"Audit-ready transaction journal with running balance"
11	CSV Export	"Data export for finance teams and external systems"

8. Key Metrics & Outcomes

Metric	Value
Properties managed	8+ villas/buildings
Apartments/units tracked	8+ per property
Expense categories	20+
Report types	4 (Balance, Ledger, Monthly, Category)
Split allocation modes	4 (All equal, Single, Selected equal, Custom)
User roles	3 (General Manager, Villa Manager, Viewer)
Languages	2 (English, Arabic)
API endpoints	40+ REST endpoints
Export formats	CSV (all finance modules)

9. Conclusion

Villa Manager Pro demonstrates end-to-end delivery of a **finance planning and budgeting platform** using the **complete Panasonic Phase II technology stack**:

.NET Core 8.0 · Angular 19 + DevExtreme · SQL Server 2022 · Azure Cloud + CI/CD · Power Apps

The solution implements **business plan analysis, cost budgeting, revenue tracking, cost allocation, forecasting via recurring templates, operating profit estimation, and multi-entity reporting** — directly aligned with Panasonic MEA's Finance Planning & Budgeting application requirements.

We bring both **technical depth** (full-stack delivery across all required technologies) and **functional understanding** (finance planning workflows, allocation logic, stakeholder reporting) — addressing the exact gap Panasonic identified with their previous implementation partner.

Appendix: Screenshot Index

#	Screenshot	File	Module
1	Login Screen	login-screen.png	Authentication
2	Finance Dashboard	dashboard.png	Dashboard / Business Plan Analysis
3	Switch Property	switch-property.png	Multi-Entity Management
4	Apartment Statement	apartment-statement.png	Unit P&L / Cost Center

#	Screenshot	File	Module
5	Apartments List	apartments-list.png	Cost Center Master Data
6	Reports — Full Ledger	reports-ledger.png	Financial Reporting
7	Reports — Monthly	reports-monthly.png	Period Analysis / Forecasting
8	Expenses List	expenses-list.png	Cost Tracking / Actuals
9	Expense Split Logic	expense-split.png	Cost Allocation Engine
10	Recurring Templates	recurring-templates.png	Budget Automation

*Document prepared for internal team presentation and Panasonic MEA Phase II proposal.
Villa Manager Pro — Finance & Operations Management Platform*